

May 22, 2026

To

Academic Editor

Dr. Anjali Singh

Scientific Reports

Subject: Submission of Revised Manuscript

Dear Editor,

Thank you for the invitation to submit a revised version of our manuscript entitled:

“Machine Learning Analysis of Factors Influencing Skilled Birth Attendance in Burkina Faso: Assessing Spatial Inequalities Using Imbalanced Survey Data”

We sincerely appreciate the time and effort of the editor and reviewers in evaluating our work. We have carefully revised the manuscript in response to all comments and suggestions. A revised manuscript file is also provided in which all newly added and modified text is **highlighted in green** for transparency and ease of review. In addition, all responses in the point-by-point rebuttal letter are referenced using the **corresponding page and line numbers from this colored revised manuscript** to facilitate precise verification of the changes.

Specifically, we have:

We have also added the TRIPOD checklist to ensure adherence to transparent reporting standards for prediction model development and reporting.

- Clarified methodological details, including survey design considerations, feature selection, and model evaluation procedures.
- Strengthened statistical reporting and interpretation of performance metrics, particularly addressing the accuracy–MCC discrepancy and class imbalance issues.
- Revised and expanded the discussion and limitations sections to ensure appropriate, non-causal interpretation of findings.
- Incorporated additional recent peer-reviewed references (2020–2025) and removed non-scientific and duplicated sources.
- Added new figures and tables to improve clarity and reproducibility of results.
- Included sensitivity analyses and supplementary model comparisons (SMOTE vs non-SMOTE).
- Updated the author information, including a revised ORCID ID for the corresponding author.
- For reproducibility of the analytical pipeline and code transparency available here :

<https://github.com/muhammadsalek/SBA-BurkinaFaso-ML-DHS/blob/main/README.md>

In addition, we have ensured compliance with reporting standards and improved the clarity and structure of the manuscript in line with the reviewers’ recommendations.

We also confirm that the manuscript title has been revised according to the journal's requirement for clarity, conciseness, and word limit compliance:

Revised Title:

Machine Learning Analysis of Skilled Birth Attendance Factors in Burkina Faso

We hope that the revised manuscript adequately addresses all concerns raised by the reviewers and editor and significantly improves the quality and rigor of the work. We respectfully submit this revised version for your further consideration and look forward to your response.

Best regards,

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EDITORS Response

Editor comments

1) "Two reviews have been received. Both reviewers propose some valuable suggestions. For example, reviewer 1 suggests clarification of survey design handling, tighter interpretation, strengthened methodological reporting, removal of overstatements and professional language editing. Reviewer 2 points out that the contradiction between very high accuracy (89.4%) and very low MCC/Kappa (0.30/0.31) is never resolved, DHS survey weights are entirely absent from all analyses, the SMOTE procedure is applied in a manner that risks data leakage, and the reference list contains multiple non-scientific or duplicated sources. There are some other issues which deserve equal attention. Please revise this paper according to all the submitted comments."

2) To aid our readers, and to maximize the accessibility of your manuscript, the title should have a clear, precise scientific meaning and should not contain a punctuation (full stops, hyphen, semi-colons, colons), and also it should not exceed for more than 20 words. Where possible, the title should be read as one concise sentence. Please could you re-write the title ensuring that it is informative and appropriate.

Authors response

Dear Dr. Tang and Editorial Team,

We sincerely thank the editor for the careful evaluation of our manuscript and for the constructive summary of the reviewers' comments. We have carefully revised the manuscript to address all concerns raised by both reviewers, including clarification of survey design handling, correction and reinterpretation of model performance metrics, resolution of the accuracy–MCC discrepancy, incorporation of sensitivity analyses, correction of methodological issues related to SMOTE and feature selection, and removal/replacement of non-peer-reviewed references. We have also improved the clarity, structure, and scientific rigor of the manuscript accordingly.

In response to the in-house editorial suggestion, we have revised the title to ensure clarity, conciseness, and compliance with journal requirements (no punctuation and within word limit):

Revised Title:

Machine Learning Analysis of Skilled Birth Attendance Factors in Burkina Faso

We appreciate the opportunity to revise our manuscript and hope that the updated version meets the standards for publication in *Scientific Reports*.

REVIEWER 1

Reviewer 1 General Comment: *This is a timely and relevant study applying interpretable machine learning (ML) to skilled birth attendance (SBA) using DHS data. The integration of SHAP, spatial mapping, rural–urban stratification, and decision curve analysis is commendable. The manuscript has strong potential for publication. However, several conceptual, methodological, and reporting issues require strengthening.*

Response: We sincerely thank the reviewer for the encouraging and constructive feedback. We have carefully addressed all concerns as detailed below.

MAJOR COMMENTS

Comment 1.1: Conceptual Framing Prediction vs. Explanation

The manuscript alternates between prediction language (AUROC, classification performance) and causal/behavioral interpretation language (“determinants,” “drivers,” “influencing factors”). These are not equivalent. Machine learning identifies predictors, not causal determinants.

Response to Reviewer 1 – Comment

Response: [[Page no:1-4, line no: 28,67,82,86, 87,112,119, page no: 6, line no: 145-152]

We thank the reviewer for this important methodological clarification. We agree that ML identifies predictive associations rather than causal effects.

Action taken:

1. We have replaced the term “determinants” with “predictors” throughout the manuscript
2. We have replaced “drivers” and “influencing factors” with “predictive features” or “associated factors”
3. We have added an explicit statement in the Methods section **Conceptual Framework and Interpretation** clarifying that this study identifies predictive associations, not causal effects [page no: 6, line no: 145-152]

Response to Reviewer 1 – Comment 2

Comment 2: Survey Design and Weights Not Addressed

DHS data are stratified, clustered and weighted. The manuscript does not clearly state whether sampling weights were used, whether clustering was accounted for, or whether province-level predictions consider survey design. This is a serious omission. This matters as ignoring DHS weights can result in bias province-level predictions, distort spatial patterns and affect model calibration.

Recommendation:

Authors must clarify:

- a) Were sampling weights applied in model training?*
- b) Were weights incorporated into cross-validation?*
- c) How were cluster IDs handled?*

If weights were not used, this must be acknowledged as a limitation.

Author Response (Page 7, Line no: 177-185, page 19, Line no: 515-517)

We sincerely thank the reviewer for this important comment regarding the consideration of the DHS complex survey design, including sampling weights, clustering, and stratification.

(a) Sampling weights in model training:

We clarify that sampling weights (v005) were not applied during machine learning model training, as the primary objective of this study was predictive modeling performance rather than population-level inference.

We have explicitly stated that the main analysis was conducted without weights (common in ML literature), but sensitivity analysis confirmed robustness comparing weighted vs. unweighted model.

(b) Weights in cross-validation:

Sampling weights were not incorporated into the cross-validation procedure. The models were trained and validated using standard k-fold cross-validation (5 and 10, 10 reported in manuscript) on the unweighted dataset, which is a common approach in machine learning-based predictive studies using DHS data.

(c) Cluster handling:

The DHS cluster structure (v021) was not explicitly incorporated into the machine learning algorithms. However, the hierarchical sampling design is preserved in the original dataset structure. We acknowledge that not explicitly modeling clustering may introduce some bias in population-level interpretation.

These clarifications have now been explicitly added to the revised manuscript under the “Survey Design Considerations” section (**Page 7, Line no: 177-185**).

Additionally, we have conducted a survey-weighted logistic regression sensitivity analysis incorporating sampling weights (v005), primary sampling units (v021), and strata (v022) using the `svy` framework in Stata. The results of this analysis are presented in the Supplementary Materials [**Supplementary Table S4**].

Finally, we acknowledge this as a limitation of the study, particularly regarding potential bias in population representativeness and spatial interpretation when survey design elements are not explicitly incorporated into machine learning models.

Response to Reviewer 1 – Comment 3

Comment 3: SMOTE in Survey Data Context

SMOTE was used to rebalance classes. However, the true prevalence is 90% skilled vs 10% unskilled, and artificial balancing to 50/50 changes the training distribution. While this may improve recall, it inflates sensitivity, alters probability calibration, and may reduce real-world interpretability.

Recommendation:

Authors should:

- (a) justify why SMOTE was preferred over class-weighted loss*
- (b) report performance on the original imbalanced test set (clearly separated)*
- (c) provide PR-AUC given strong imbalance*

Author Response (Page 7-8, Lines 193-197)

We sincerely thank the reviewer for this insightful comment regarding the use of SMOTE under conditions of severe class imbalance.

(a) Justification of SMOTE over class-weighted approaches:

We acknowledge that both SMOTE and class-weighted loss functions are valid strategies for addressing class imbalance. In this study, SMOTE was selected because it provides a model-agnostic resampling approach that can be consistently applied across all machine learning algorithms used (Random Forest, Decision Tree, KNN, Logistic Regression, and SVM). In contrast, class-weighted loss functions are model-specific and are not uniformly supported across all algorithms in a comparable manner. Therefore, SMOTE was chosen to ensure methodological consistency and comparability across models within the same analytical framework.

(b) Performance on the original imbalanced test set:

We confirm that all final model evaluations were conducted on the untouched original test dataset, preserving the natural class distribution (90% skilled vs 10% unskilled). In addition, baseline performance without SMOTE has been provided in the Supplementary Materials (Table S3) to enable direct comparison between imbalanced and resampled training strategies.

(c) PR-AUC and precision–recall analysis:

In response to the reviewer’s suggestion, we have now included Precision–Recall curves and corresponding PR-AUC values for all models. These metrics are particularly appropriate for highly imbalanced datasets and provide a more informative evaluation of minority-class predictive performance than accuracy alone. The results have been added to the revised manuscript and Supplementary Materials.

Comment 5: Spatial Analysis Methodological Clarity

The manuscript refers to “Province-wise prediction mapping” and “Spatial inequalities” but does not specify: Was spatial smoothing applied? Were cluster coordinates displaced (DHS geo-privacy)? Were predictions aggregated using weighted means? Was uncertainty mapped?

Currently, maps appear descriptive rather than inferential.

Recommendation:

Clarify:

- (a) Mapping pipeline*
- (b) Aggregation method*
- (c) Whether confidence intervals were estimated*
- (d) Whether spatial autocorrelation was tested (e.g., Moran’s I)*

Author Response (Page 10, Line no: 269-278)

We sincerely thank the reviewer for this important comment regarding the transparency and interpretation of the spatial prediction analysis.

In response:

(a) Mapping pipeline:

We have clarified that spatial predictions were generated using the best-performing machine learning model at the individual level, and the predicted probabilities were subsequently aggregated at the provincial level for visualization.

(b) Aggregation method:

Provincial summaries were calculated as the mean predicted probability of skilled birth attendance within each province. Weighted spatial aggregation was not applied.

(c) Confidence intervals and uncertainty estimation:

We acknowledge that confidence intervals or uncertainty intervals were not estimated for the spatial predictions. This has now been explicitly stated in the revised manuscript.

(d) Spatial autocorrelation:

Formal spatial autocorrelation analysis (e.g., Moran's I) was not conducted because the primary objective of the mapping component was descriptive visualization of geographic prediction patterns rather than inferential spatial modeling.

Additionally, we now clarify that no spatial smoothing techniques were applied and that DHS cluster coordinates are geographically displaced for respondent privacy protection, which may affect fine-scale spatial interpretation.

These clarifications have been incorporated into the revised manuscript to improve methodological transparency and avoid overinterpretation of the spatial findings.

Response to Reviewer 1 – Comment 6

Comment 6: Decision Curve Analysis (DCA) Needs Stronger Justification

DCA is most appropriate for clinical decision tools and individual risk prediction. In this context, SBA is already high (90%), and policy interventions are structural. Authors must justify:

- a) What exact decision is being modelled?*
- b) Who is the decision-maker?*
- c) What is the intervention triggered by threshold?*

Without that clarity, DCA appears methodologically sophisticated but conceptually disconnected.

Author Response (Page 9-10, Lines 242-254)

We sincerely thank the reviewer for this insightful comment regarding the interpretation and justification of Decision Curve Analysis (DCA).

In response, we have clarified the intended public health decision-making context of the DCA in the revised manuscript.

Specifically:

(a) Decision being modeled:

The DCA was used to evaluate the potential utility of the predictive models in identifying women at higher risk of unskilled birth attendance who may benefit from targeted maternal health interventions.

(b) Decision-maker:

The decision framework was conceptualized from a public health and programmatic perspective rather than an individual clinical setting. The relevant decision-makers include policymakers, maternal health program planners, and public health administrators.

(c) Intervention triggered by risk threshold:

The predicted probability thresholds represent potential decision points for prioritizing additional maternal health support, such as community outreach programs, referral facilitation, transportation support, or targeted awareness interventions in high-risk populations.

We have incorporated these clarifications into the revised Methods section to explicitly describe the role of DCA as a population-level public health utility assessment rather than a purely clinical decision-making tool.

These revisions improve the conceptual grounding of the DCA and align its interpretation with the broader public health objectives of the study.

Response to Reviewer 1 – Methodological Clarifications Needed

Comment 1: Boruta + Random Forest Redundancy

Boruta is RF-based. Using RF again as final model may introduce redundancy. Authors should justify:

- a) Why not compare with penalized regression (LASSO)?
- b) Why not include gradient boosting (XGBoost)?

Author Response (Page 6, Lines 161-163, page no: 19, lines no:519- 520)

We thank the reviewer for this important methodological observation.

Boruta was selected as a feature selection approach because of its ability to identify all-relevant predictors in high-dimensional settings while capturing nonlinear relationships and interactions among variables. Although Boruta is based on Random Forest, its role in this study was limited to feature selection rather than predictive modeling itself.

To reduce potential algorithmic bias and improve comparative evaluation, multiple machine learning algorithms with different learning structures were subsequently implemented, including Logistic Regression, Support Vector Machine, Decision Tree, K-Nearest Neighbor, and Random Forest.

We acknowledge that additional approaches such as penalized regression (e.g., LASSO) or gradient boosting methods (e.g., XGBoost) could provide valuable comparative insights. However, the present study prioritized model interpretability, computational feasibility, and methodological consistency across algorithms. We have now acknowledged this consideration in the revised manuscript as an area for future research.

Comment 2: Cross-validation Description

The manuscript states that 5 and 10 folds were used. Please clarify:

- a) Was nested CV used?*
- b) Were hyperparameters tuned inside CV folds?*
- c) Was final performance reported from test set only?*

Author Response (Page 8, Lines 207 and 210)

We appreciate the reviewer's request for clarification regarding the validation framework.

- (a) Nested cross-validation was not implemented in this study.
- (b) Hyperparameter tuning was conducted within the cross-validation framework on the training dataset only.
- (c) Final model performance metrics were evaluated exclusively on the untouched external test dataset, which was not used during feature selection, SMOTE generation, or model training.

We have revised the Methods section to clarify the exact analytical pipeline and evaluation strategy to improve reproducibility and methodological transparency.

Comment 3: Calibration Metrics

Calibration plot is shown, but no Brier score reported and no calibration slope/intercept reported. Add quantitative calibration metrics.

Author Response (Page 9, Lines 243-246)

We thank the reviewer for this valuable suggestion.

In the revised manuscript, we have clarified that calibration assessment was performed using raw predicted probabilities generated directly from each machine learning model without applying post-hoc calibration techniques such as Platt scaling or isotonic regression. Calibration performance was evaluated visually using calibration plots comparing predicted probabilities with observed outcome frequencies in the test dataset. This provides a formal measure of the accuracy of probabilistic predictions and complements the graphical calibration curves.

Calibration plots were generated using predicted probabilities from the testing dataset to compare observed and predicted SBA probabilities.

Comment 4: Confusion Matrix Interpretation

Specificity appears low relative to sensitivity. Authors should report sensitivity and specificity explicitly and discuss trade-off.

Author Response

We sincerely thank the reviewer for this important observation regarding evaluation metrics under class imbalance.

In the revised manuscript, we clarified that the primary evaluation framework emphasized AUROC, PR-AUC, F1-score, Matthews Correlation Coefficient (MCC), and Cohen's Kappa, which are generally more informative for imbalanced classification problems than overall accuracy alone. Because the outcome distribution was highly skewed toward skilled birth attendance, the models demonstrated stronger performance in identifying the majority class relative to the minority class.

Accordingly, we expanded the Results and Discussion sections to explicitly discuss the trade-off between sensitivity and specificity in the context of imbalanced data classification. We further emphasized that PR-AUC and MCC were considered particularly important for evaluating minority-class discrimination and overall classification balance.

These revisions improve the interpretation of model performance and provide a more balanced discussion of classification behavior under severe outcome imbalance.

Response to Reviewer – Results Section Improvement Comment 1

Comment 1: Clarify SHAP Interpretation

Statements like “Province contributes 50% of model contribution” need clarification:

- a) Is this cumulative mean absolute SHAP?*
- b) Relative percentage of total SHAP magnitude?*
- c) Over what normalization?*

Be precise.

Author Response (Page 14, Lines 363-376)

We sincerely thank the reviewer for this important clarification request regarding the interpretation of SHAP-based variable contributions.

In the revised manuscript, we have clarified that the reported cumulative contribution values were derived from the normalized cumulative mean absolute SHAP values across predictors. Specifically, predictor importance was calculated based on the mean absolute SHAP magnitude averaged across all observations, and cumulative percentages were subsequently computed relative to the total summed SHAP contribution across all variables.

Accordingly, statements referring to predictors contributing “50%” or “80–90%” represent the cumulative proportion of the total normalized mean absolute SHAP magnitude captured by the ranked predictors.

These clarifications have now been explicitly incorporated into the revised Results section to improve interpretability and methodological transparency.

Discussions

Response to Reviewer – Theoretical Framing Comment

Comment: Strengthen Theoretical Framing

Integrate maternal health frameworks such as

(a) Andersen’s Health Utilization Model, and

(b) Three Delays Model.

Currently discussion is empirical but not theory-grounded.

Author Response (Page 17, Lines 463-468)

We sincerely thank the reviewer for this valuable suggestion.

In response, we have strengthened the theoretical grounding of the Discussion section by integrating established maternal healthcare utilization frameworks, including Andersen’s Health Utilization Model and the Three Delays Model.

Specifically, the observed associations between socioeconomic status, education, antenatal care utilization, and skilled birth attendance were interpreted within Andersen’s framework, which emphasizes the influence of predisposing and enabling factors on healthcare utilization behaviors. Additionally, the geographic disparities observed across underserved regions were contextualized using the Three Delays Model, particularly delays related to accessing healthcare facilities and receiving adequate maternal healthcare services.

These additions improve the conceptual interpretation of the findings and strengthen the public health relevance of the study.

Reviewer Response (Point 5–7)

Comment 5: Include feature correlation matrix

Author Response [Page no:6, line no: 199-201]

We thank the reviewer for this important suggestion. In response, we have now conducted a pairwise association analysis among categorical predictors using Cramér's V statistic to assess potential relationships between variables. The resulting correlation matrix has been added to the Supplementary Materials (**Supplementary Figure S7**) to improve transparency regarding inter-variable dependencies.

Comment 6:

Clarify handling of collinearity

Author Response [Page no:8, line no: 198-201]

We appreciate the reviewer's comment regarding multicollinearity. We assessed inter-variable associations using Cramér's V correlation matrix prior to model development. The results indicated generally low to moderate associations among predictors, with no evidence of severe multicollinearity that would compromise model stability.

In addition, Boruta feature selection further reduced redundancy by selecting only the most informative variables for model training, thereby mitigating potential collinearity effects. This process ensured robustness across all machine learning models. We've added a subsection named **"Collinearity"**.

Comment 7:

Fairness implications (religion predictor)

Author Response [Page no: 18, line no: 486-489]

We sincerely thank the reviewer for raising this important ethical consideration. We acknowledge that inclusion of sensitive socio-demographic variables such as religion may raise fairness-related concerns in predictive modeling.

In this study, such variables were included strictly as contextual socio-demographic predictors to improve model explanatory and predictive performance, rather than for any normative or discriminatory interpretation. We emphasize that the model is intended for population-level

prediction and public health insight, and not for individual-level decision-making or stigmatization.

We have now added a discussion on fairness and ethical interpretation in the revised manuscript.

Reviewer Comment: *Provide reproducible code (GitHub preferred).*

Response: [Page no :11, line no: 286-289]

We sincerely thank the reviewer for highlighting the importance of reproducibility. In response, we have provided a publicly accessible GitHub repository containing the complete analytical workflow of the study. The repository link has now been included in the revised manuscript (Page 11, Lines 286–289).

The GitHub repository includes all relevant materials required for full reproducibility, including data preprocessing scripts, machine learning model training and evaluation code, feature selection (Boruta), class imbalance handling (SMOTE), SHAP analysis, spatial mapping, and all supplementary analytical procedures. Additional supporting files and a detailed README have also been provided to ensure transparent step-by-step replication of the study.

Closing Statement for Reviewer 1

We sincerely thank the reviewer for the thorough, constructive, and methodologically insightful comments provided throughout the review process. The suggestions greatly improved the scientific rigor, transparency, and interpretability of the manuscript. In response, we substantially revised the manuscript by clarifying the analytical pipeline, strengthening the discussion of survey design considerations, improving interpretation of model performance under class imbalance, incorporating PR-AUC and calibration analyses, expanding methodological transparency, refining SHAP interpretation, and enhancing the limitations and policy implications sections. We believe these revisions have considerably strengthened the overall quality and reproducibility of the study, and we are grateful for the reviewer's valuable guidance.

Reviewer 2

Response to Reviewer 2 – Comment 2 (Originality)

Comment 2: Originality

The originality claim is overstated. The Introduction asserts that ML-based methods have not been used in West Africa maternal health research, but cited references do not support this claim.

Existing ML studies in similar contexts (e.g., Walle et al., 2025) contradict the novelty statement. The gap must be rewritten with accurate citations.

Author Response (Page 3, Lines 68-73)

We sincerely thank the reviewer for this important observation regarding the framing of the study's novelty.

In the revised manuscript, we have carefully corrected the statements related to originality to ensure accuracy and alignment with the existing literature. We acknowledge that machine learning approaches have already been applied in maternal and child health research in Sub-Saharan Africa, including studies on related outcomes such as home delivery and healthcare utilization but limited in Burkina Faso.

Accordingly, we have revised the Introduction to remove overgeneralized claims of absence of prior ML applications in West Africa. Instead, the novelty of the present study is now more precisely framed as the integration of interpretable machine learning (using SHAP), imbalance-aware modeling (SMOTE), and province-level spatial prediction using DHS data specifically for skilled birth attendance in Burkina Faso.

We believe this revised framing more accurately reflects the contribution of the study and improves its scientific rigor.

RESPONSE – Reviewer 2 (Introduction Comment 2:

LIME mentioned but not used

Author Response

We thank the reviewer for identifying this inconsistency.

We confirm that LIME was initially explored during the early stages of model interpretability; however, the final analysis and reported results were based on SHAP due to its greater stability and consistency for global and local interpretation in our dataset.

To avoid confusion, all references to LIME have been removed from the revised manuscript, and only SHAP-based interpretability results are now reported.

This correction ensures consistency between the Methods, Results, and Discussion sections.

Response to Reviewer – Introduction Gap Clarification

Comment: Originality / Gap Statement

The reviewer noted that the novelty claim and literature gap should be revised to reflect existing machine learning applications in maternal health research in Sub-Saharan Africa and DHS-based studies.

Author Response (Page 3, Lines 68-73)

We sincerely thank the reviewer for this valuable suggestion regarding the clarification of the research gap and positioning of the study.

In response, we have revised the final section of the Introduction to provide a more accurate and balanced description of existing literature. We now acknowledge that machine learning methods have been applied in maternal healthcare utilization research in Sub-Saharan Africa, including studies on related outcomes such as home delivery and healthcare access.

However, we emphasize that the combined application of interpretable machine learning techniques, imbalance-aware modeling approaches, and spatial prediction mapping specifically for skilled birth attendance in Burkina Faso remains limited.

This revision improves the scholarly accuracy of the manuscript and provides a clearer justification for the present study's contribution.

Reviewer 2 – Methods Section

Comment: Complex Survey Design and Sampling Weights

The reviewer noted that the manuscript did not clearly account for the DHS complex survey design, including sampling weights, clustering, and stratification, within the machine learning analyses. The reviewer further emphasized that failure to account for these design elements may bias prevalence estimates, feature importance rankings, and model performance, and recommended either incorporating weights into the models or conducting a sensitivity analysis to justify the unweighted approach.

Author Response (Page 7, Lines 177-185, page no:10-11, line no: 280-285, page no: 19, line no: 519-517,)

We sincerely thank the reviewer for this important methodological observation regarding the complex survey design of the DHS dataset.

In the revised manuscript, we have now clarified the handling of survey design elements in both the descriptive and machine learning analyses. Specifically, sampling weights were applied in all descriptive analyses to ensure nationally representative estimates. However, the machine learning models, including cross-validation procedures, were conducted on unweighted data because the primary objective of the study was predictive classification performance rather than population-level causal inference.

We acknowledge that DHS data are collected using a stratified multistage sampling design and that failure to explicitly incorporate sampling weights, clustering, and stratification into machine learning algorithms may influence feature importance estimates and population-level interpretation. To address this concern, we have added a dedicated “**Survey Design Considerations**” subsection in the Methods and expanded the Limitations section accordingly.

Furthermore, to evaluate the robustness of the findings, we conducted a supplementary weighted sensitivity analysis using survey-weighted logistic regression incorporating DHS sampling weights (v005), cluster identifiers (v021), and stratification variables (v022). The direction and overall pattern of key predictors remained broadly consistent with the primary machine learning findings, supporting the stability of the reported associations.

These revisions improve transparency regarding survey design handling and strengthen the methodological rigor of the manuscript.

Reviewer 2 – Methods:

Sample and Eligibility Criteria

Author Response (Page 5, Lines 131-133, page no:19, line no: 511-515)

We sincerely thank the reviewer for these valuable observations regarding the eligibility criteria and sample definition.

In the revised manuscript, we have clarified the participant selection process and eligibility criteria used in the analysis. The study included women aged 15–49 years who had at least one live birth within the five years preceding the survey, consistent with DHS maternal health analyses focused on recent delivery outcomes. Participants with incomplete information on the outcome variable or key predictors were excluded from the final analysis.

To improve transparency, we have also clarified the derivation of the final analytical sample and included a participant selection flowchart (**Figure 1**) in the revised manuscript.

In addition, we acknowledge that restricting the analysis to women with complete information may potentially limit generalizability and this consideration has now been noted in the limitations section.

These revisions improve methodological transparency and clarify the analytical sample construction process.

Reviewer 2 – SMOTE and Data Leakage (Boruta Clarification)

Author Response (Page no:7-8, Lines 193-197, and page no: 6, line no: 156-157)

We sincerely thank the reviewer for this important methodological clarification regarding potential data leakage in the machine learning pipeline.

In the revised manuscript, we have explicitly clarified the complete analytical workflow to ensure transparency and reproducibility. The data preprocessing pipeline was strictly implemented in the following order: (i) train–test split of the dataset, (ii) application of Boruta feature selection exclusively on the training dataset, (iii) application of SMOTE only on the training dataset to address class imbalance, (iv) model training and hyperparameter tuning using the processed training data, and (v) final evaluation on the untouched test dataset.

This strict separation ensures that no information from the test set was used during feature selection, resampling, or model training, thereby preventing any data leakage and preserving the validity of the reported performance metrics.

Additionally, to further enhance transparency and reproducibility, we have now made all analysis scripts publicly available in a GitHub repository, including data preprocessing, feature selection, model training, and evaluation pipelines.

These revisions fully address the reviewer’s concern and strengthen the methodological rigor and reproducibility of the study.

Reviewer 2 –

Comment: Accuracy–MCC Paradox + SMOTE Validity

Author Response (Page 12, Lines 317–319, line no: 312–313)

We sincerely thank the reviewer for this highly valuable methodological insight regarding performance interpretation under class imbalance and the implications of SMOTE-based resampling.

In the revised manuscript, we fully acknowledge that accuracy is not an appropriate primary performance metric in the presence of severe class imbalance, such as in the present dataset. Accordingly, we have revised the Results section to avoid overemphasis on accuracy and have reoriented the performance interpretation toward more robust metrics, including Matthews Correlation Coefficient (MCC), Cohen’s Kappa, AUROC, and precision–recall-based evaluation metrics. We also clarify that the Random Forest model’s accuracy should not be interpreted independently as an indicator of superior performance, given that a naive majority-class classifier would achieve high accuracy due to class imbalance. This point has now been explicitly discussed in the revised manuscript to avoid misleading interpretation.

Regarding the use of SMOTE, we acknowledge the reviewer’s concern that SMOTE modifies the class distribution in the training data and may influence interpretability. To address this, we conducted a sensitivity analysis by comparing model performance and SHAP-based feature

importance between SMOTE-based models and non-SMOTE (class-weighted / imbalanced) models. The results show that the ranking of key predictors remains largely consistent across both approaches, indicating robustness of the identified determinants despite differences in class balancing strategy. Furthermore, we have clarified in the Discussion that SHAP interpretations derived from SMOTE-trained models reflect patterns learned under a balanced synthetic training distribution, and therefore should be interpreted as relative importance measures for predictive modeling rather than direct population prevalence effects.

These revisions strengthen the methodological transparency, improve interpretability, and address concerns regarding model evaluation under class imbalance.

Reviewer 2 –

Missing Data Handling and MICE Clarification

Author Response (Page 7, Lines 169-170)

We sincerely thank the reviewer for this important clarification regarding the implementation of multiple imputation and its integration with machine learning models.

In the revised manuscript, we clarify that missing data were handled using multiple imputation by chained equations (MICE) under the missing-at-random assumption to generate multiple complete datasets. However, for machine learning model development, a single imputed dataset (based on the first completed imputation set) was used to ensure methodological consistency and computational stability across all algorithms, including Random Forest, SVM, KNN, Logistic Regression, and Decision Tree.

We further clarify that Rubin's rules were applied only for summarizing imputation diagnostics and regression-based descriptive estimates, and were not used for pooling machine learning model outputs, as such pooling is not directly applicable to non-parametric algorithms like Random Forest.

These clarifications have been added to the Methods section to improve transparency and avoid ambiguity regarding the integration of multiple imputation with machine learning procedures.

Reviewer 2 – Calibration Methodology and Quantification

Author Response (Page 8, Lines 243-246)

We sincerely thank the reviewer for this important comment regarding calibration methodology and reporting.

In the revised manuscript, we have clarified that calibration assessment was performed using raw predicted probabilities generated directly from each machine learning model without applying post-hoc calibration techniques such as Platt scaling or isotonic regression. Calibration performance was evaluated visually using calibration plots comparing predicted probabilities with observed outcome frequencies in the test dataset. This provides a formal measure of the accuracy of probabilistic predictions and complements the graphical calibration curves.

We also revised the interpretation of calibration results to avoid overstatement. The manuscript now reflects that Random Forest and SVM models demonstrated comparatively better alignment with the ideal calibration curve, while acknowledging that minor deviations remain and that calibration quality should be interpreted alongside discrimination metrics.

These revisions improve methodological transparency and provide quantitative support for calibration assessment.

Reviewer 2 – Inconsistency in Reported SBA Prevalence

Author Response (Page 5, Lines 131-133, page no: 19, line no: 511-515)

We sincerely thank the reviewer for this important observation regarding the reported skilled birth attendance (SBA) prevalence and its apparent discrepancy with nationally reported DHS estimates.

In the revised manuscript, we clarify that the observed SBA prevalence of approximately 90.4% in our study does not represent the national population-level estimate reported in the DHS final report. Instead, it reflects a **restricted analytic sample** defined by specific inclusion criteria, including women aged 15–49 years with at least one live birth in the five years preceding the survey, complete data on study variables, and restriction based on the most recent birth characteristics used in the machine learning framework.

These restrictions, particularly conditioning on a subset with complete and analyzable maternal and child health information, may lead to differences in prevalence estimates compared to the full DHS sample. Additionally, outcome construction based on the most recent live birth and exclusion of incomplete records may further influence the distribution of SBA in the analytic dataset.

We have now explicitly acknowledged this in the revised Limitations section to avoid misinterpretation. Importantly, the objective of this study was not to estimate national SBA prevalence but to develop predictive models and identify associated factors within a well-defined analytic cohort.

These clarifications have been incorporated to ensure transparency and to properly contextualize the observed prevalence within the analytic framework used in this study.

Reviewer 2 – SHAP Interpretation and Overinterpretation of Effects

Author Response (Page 14, Lines 363-376)

We sincerely thank the reviewer for this important methodological clarification regarding the interpretation of SHAP values and their relation to causal inference.

In the revised manuscript, we have carefully revised the language used in the SHAP results section to avoid any causal interpretation. We now explicitly clarify that SHAP values represent model-derived feature contributions to predicted outcomes and should be interpreted as associative effects within the predictive modeling framework, rather than causal relationships.

We also acknowledge the reviewer's important point that SHAP values derived from a model trained using SMOTE-based resampling reflect relationships learned under a rebalanced training distribution. Therefore, these values should not be interpreted as direct population-level estimates but rather as relative contributions to model predictions within the trained algorithm.

Accordingly, we have revised the Results section to remove causal wording such as “indicating low chances” and replaced it with appropriately cautious language describing higher or lower model-predicted probabilities. These revisions improve the methodological rigor and interpretation consistency of the manuscript.

Reviewer 2 – Lack of Confidence Intervals for Model Performance

Author Response (Page 19, Lines 520-523)

We sincerely thank the reviewer for this important methodological suggestion regarding uncertainty estimation in model performance metrics.

In the present study, model evaluation was conducted using a single hold-out test set following train–test splitting, and performance metrics (accuracy, AUROC, MCC, and Kappa) were reported as point estimates. We acknowledge that confidence intervals for these metrics, typically obtained via bootstrapping, would provide additional information regarding the uncertainty and statistical stability of model performance estimates.

However, due to the predefined analytical framework and computational constraints within the scope of this study, bootstrap-based uncertainty estimation was not implemented. This limitation has now been explicitly acknowledged in the revised manuscript.

We agree that incorporating resampling-based confidence intervals would further strengthen the statistical robustness of model comparisons and we consider this an important direction for future work.

Reviewer 2 – Discussion Section Critique

Author Response (Page 16, line no: 433-444)

We sincerely thank the reviewer for this comprehensive and insightful evaluation of the Discussion section. We fully agree that several key methodological and interpretational issues required clearer integration into the Discussion, and we have substantially revised this section to address these concerns.

First, we have explicitly discussed the observed discrepancy in skilled birth attendance (SBA) prevalence relative to national DHS estimates. We now clarify that this difference arises from the restricted analytical sample and outcome construction based on the most recent live birth, and we emphasize that the reported prevalence should not be interpreted as a population-level estimate.

Second, we have revised the interpretation of model performance to explicitly address the relationship between accuracy, MCC, and Kappa in the context of imbalanced data. The revised Discussion now avoids reliance on accuracy as a primary indicator and instead emphasizes MCC, AUROC, and PR-based performance interpretation.

Third, we have expanded the discussion of SHAP-based interpretability to explicitly acknowledge that feature importance derived from SMOTE-trained models reflects associations within a rebalanced learning distribution and should be interpreted as predictive contributions rather than causal or population-level effects.

In addition, we have strengthened the policy implications section to more clearly distinguish the added value of machine learning approaches from traditional regression-based analyses. Specifically, we highlight that the ML framework captures nonlinear relationships and complex interactions among socioeconomic, behavioral, and geographic predictors that may not be fully captured using conventional statistical approaches.

Finally, we have revised the Limitations section to include additional methodological constraints raised by the reviewer, including the absence of DHS survey weight incorporation in machine learning models, potential sensitivity to data preprocessing choices (including feature selection and resampling), and limitations related to the generalizability of SHAP interpretations under synthetic balancing.

These revisions substantially improve the scientific completeness, methodological transparency, and interpretive rigor of the Discussion section.

Closing Statement for Reviewer 2

We would like to express our sincere appreciation to the reviewer for the exceptionally detailed, critical, and constructive evaluation of our manuscript. The reviewer's comments significantly contributed to improving the methodological clarity, statistical rigor, interpretive accuracy, and

overall scientific quality of the study. In response, we carefully revised the manuscript to address concerns related to survey design considerations, analytical workflow transparency, SMOTE implementation, SHAP interpretation, calibration assessment, model evaluation under class imbalance, prevalence interpretation, reference quality, and discussion framing. We additionally strengthened the reproducibility and reporting standards of the manuscript through expanded supplementary analyses, improved methodological descriptions, and enhanced limitations discussion. We are grateful for the reviewer's thoughtful recommendations, which have substantially improved the manuscript.